

3. Introduction

In today's fast-paced world, monitoring and controlling speed has become essential in various domains such as transportation, industry, and public safety. Over-speeding is one of the major causes of accidents and damage, especially in sensitive areas like school zones, factory floors, and residential . To address this issue, we have designed and implemented a **Speed Detector System** using basic electronic components and an Arduino Nano microcontroller.

This project aims to detect the speed of a moving object over a short fixed distance and alert the user if the speed exceeds a predefined safe limit. The system uses two **Infrared (IR) Sensors** placed 13 cm apart to detect the motion of an object. When an object passes the first sensor, the timer starts, and when it crosses the second sensor, the timer stops. Using the known distance and the recorded time, the speed is calculated with the formula:

Speed = Distance / Time

4. Components Used

Jumper Wire : A jumper wire is a short electrical wire used to connect components on a breadboard or circuit without soldering.



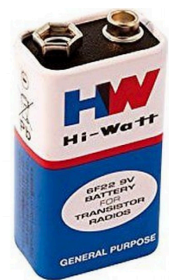
Zero PCB: A jumper wire is a short electrical wire used to connect components on a breadboard or circuit without soldering .



Capacitor : A 220 μ F 25V electrolytic capacitor Stores electric charge, filters signals, and operates safely up to 25 volts DC



Battery :The **Hi-Watt (HW) 9V battery** is a non-rechargeable zinc-carbon battery delivering 9 volts. It's widely used in devices like smoke detectors, toys, clocks, and multimeters due to its affordability and reliability.



LCD Display: A 16x2 LCD Display (Green Backlight) shows 2 lines of 16 characters each, with a green backlight for visibility.



Potentiometer : A potentiometer is a variable resistor used to adjust voltage levels, controlling brightness, volume, or other electronic parameters.



IR Sensor : An IR sensor detects infrared light from objects, used for obstacle detection, motion sensing, and distance measurement in electronics projects.



Arduino Nano : Arduino is an open-source microcontroller platform used for building electronics projects, featuring easy programming, digital/analog 10 pins. and USB interface



Buzzer : A buzzer is an audio signalling device that produces sound using vibrations, commonly used for alerts, alarms, and notifications in circuits



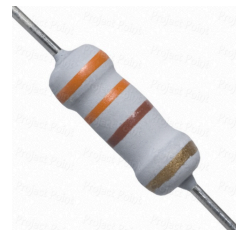
LED : An LED (Light Emitting Diode) emits light when current flows through it, used for indication. lighting, and display in electronics



USB Connector : A USB connector is used for data transfer and charging, providing a standard interface for connecting devices to computers or power sources.



Resistor : A 330-ohm resistor limits current in a circuit, commonly used for protecting LEDs and controlling current flow in electronics.



8. Working of Speed Detector System

The Speed Detector System is designed to measure the speed of an object passing between two IR sensors placed at a fixed distance of 13 cm. The system is built using various components including Jumper Wires, Zero PCB, Capacitor, Battery, LCD Display, Potentiometer, IR Sensors, Arduino Nano, Buzzer, LED, BMS, Boost Converter, and Resistors.

When an object interrupts the first IR sensor, the Arduino records the start time. As the object passes and interrupts the second IR sensor, the end time is recorded. Using the formula:

Speed = Distance / Time,

the Arduino calculates the speed of the object. The fixed distance (13 cm) is used in the formula, and the time is measured in milliseconds using the `millis ( )` function of Arduino.

The calculated speed is then compared with a predefined safe speed threshold. If the speed is within the safe limit, the LCD displays the current speed and a message "Safe Speed" in green backlight. If the speed exceeds the limit, the system activates the buzzer and lights up the red LED to alert the user. Simultaneously, the LCD displays the current speed along with a warning message "Over Speed".

The Boost Converter ensures a stable voltage supply to the system, and the BMS safely manages the rechargeable battery power. The Potentiometer is used to adjust the contrast of the LCD display for better visibility.

This project serves as a basic and effective model for speed monitoring in safety-critical zones such as school areas, factories, or automated transport systems.

10. Code

```
#include <LiquidCrystal.h>

// LCD pins: RS, EN, D4, D5, D6, D7
LiquidCrystal lcd(7, 8, 9, 10, 11, 12);

// Pin assignments
const int irSensor1 = 2;
const int irSensor2 = 3;
const int ledPin = 5;    // LED moved to D7
const int buzzerPin = 6; // Buzzer remains on D6

// Speed parameters
const float distance = 0.13; // 13 cm = 0.13 meters
const float speedLimit = 2.0; // Speed threshold in m/s

unsigned long time1 = 0;
unsigned long time2 = 0;
float speed = 0;

void setup() {
  pinMode(irSensor1, INPUT);
  pinMode(irSensor2, INPUT);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  lcd.begin(16, 2);
  lcd.print("Speed Detector");
  delay(2000);
  lcd.clear();

  Serial.begin(9600); // For debugging
```

```
}
```

```
void loop() {  
  lcd.setCursor(0, 0);  
  lcd.print("Waiting Object");  
  
  // Wait for object at IR sensor 1  
  while (digitalRead(irSensor1) == HIGH);  
  time1 = millis();  
  digitalWrite(ledPin, HIGH);  
  lcd.clear();  
  lcd.setCursor(0, 0);  
  lcd.print("Object Detected");  
  
  // Wait for object at IR sensor 2  
  while (digitalRead(irSensor2) == HIGH);  
  time2 = millis();  
  digitalWrite(ledPin, LOW);  
  
  // Calculate speed  
  float timeTaken = (time2 - time1) / 1000.0;  
  speed = distance / timeTaken;  
  
  // Debug output  
  Serial.print("Time: ");  
  Serial.print(timeTaken);  
  Serial.print(" s | Speed: ");  
  Serial.print(speed);  
  Serial.println(" m/s");  
  
  // Display on LCD  
  lcd.clear();  
  lcd.setCursor(0, 0);  
  lcd.print("Speed: ");
```

```
lcd.print(speed, 2);  
lcd.print(" m/s");
```

```
if (speed > speedLimit) {  
    Serial.println("Overspeed! Buzzer ON");  
    digitalWrite(buzzerPin, HIGH); // Buzzer ON  
    digitalWrite(ledPin, HIGH);    // LED ON  
    lcd.setCursor(0, 1);  
    lcd.print("Over Speed!");  
    delay(1200);                  // Buzzer & LED stay ON for 5s  
    digitalWrite(buzzerPin, LOW); // Buzzer OFF  
    digitalWrite(ledPin, LOW);    // LED OFF  
} else {  
    lcd.setCursor(0, 1);  
    lcd.print("Speed Normal");  
    delay(2000);  
}
```

```
lcd.clear();  
}
```

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13. Challenges Faced in Building the Speed Detector System

1. Soldering Components Neatly

Ensuring strong, clean, and reliable solder joints was challenging—especially on a compact board like the Arduino Nano. Poor soldering can lead to loose connections or short circuits.

2. Sensor Alignment & Accuracy

Positioning the IR sensors exactly 13 cm apart and perfectly aligned to detect passing objects consistently required precision.

3. Timing Precision

Accurate measurement of time between the two IR sensor triggers was critical. Minor timing errors can significantly affect speed calculation.

4. Debouncing & False Triggering

IR sensors were sensitive to ambient light and reflections, which sometimes caused false detections and inconsistent results.

5. Power Supply Stability

Maintaining a stable power supply for the sensors, buzzer, and LED was important to prevent resets or erratic behavior.

6. Code Logic & Optimisation

Writing real-time code to accurately capture input, calculate speed, and activate outputs without delay required careful planning and debugging.

7. Testing & Calibration

Multiple tests were needed with different object sizes and speeds to fine-tune the system for consistent performance.

14. Conclusion

The speed detector system using Arduino Nano, IR sensors, buzzer, and LED successfully demonstrates how embedded systems can be applied to measure the speed of a moving object over a fixed distance. By calculating the time taken to travel between two IR sensors placed 13 cm apart, the system can provide real-time speed detection and alert using visual and audio indicators.

Throughout the project, various practical challenges such as precise soldering, accurate sensor alignment, noise interference, and real-time coding were encountered and overcome. The project not only enhanced understanding of hardware interfacing and timing-based calculations but also improved hands-on skills in circuit design, debugging, and system testing.

Overall, the project serves as a simple yet effective example of how microcontrollers and sensors can be integrated to build useful, real-world applications.

15. Bibliography

1. **Arduino Official Documentation**
<https://www.arduino.cc/en/Guide/HomePage>
(For understanding Arduino Nano pin configuration and programming)
2. **IR Sensor Working and Applications** – Electronics Hub
<https://www.electronicshub.org/ir-sensor/>
(For understanding the working principle of IR sensors)
3. **Speed Measurement Using Arduino** – Circuit Digest
<https://circuitdigest.com>
(For guidance on logic and coding for speed measurement)
4. **Embedded Systems: Introduction to the Arduino Platform**
– Coursera / University of California, Irvine
(For foundational understanding of embedded programming and real-time systems)
5. **YouTube Tutorials** – Various creators (e.g., "Speed Sensor with Arduino", "IR Sensor Projects")
(For practical insights and project inspiration)
6. **Datasheets of Components**
 - Arduino Nano Datasheet
 - IR Sensor Module Datasheet
 - Buzzer and LED specifications
(For hardware characteristics and limitations)

11. Applications of the Speed Detector System

1. Object Speed Measurement in Labs

Useful in school or college physics labs to measure the speed of small objects in motion experiments.

2. Toy Car or Miniature Vehicle Speed Testing

Can be used in hobby projects or robotics competitions to track how fast a small vehicle is moving.

3. Conveyor Belt Monitoring

Helps in detecting and regulating the speed of items moving on a conveyor belt in automation systems.

4. Traffic Monitoring (Scaled-Down Models)

With some upgrades, the system could be used in prototype models to simulate and study traffic speed detection.

5. Sports Training Equipment

Can be adapted for measuring object speeds in activities like rolling balls in cricket/bowling practice setups.

6. Industrial Automation Projects

Helpful in monitoring speed in automated machinery where precise control over moving parts is required.

7. Security and Intruder Alert Systems

Can be part of a system that detects the speed of movement in restricted areas, triggering alarms for fast motion.

9. Procedure

1. Component Collection

Gather all necessary components:

- Arduino Nano
- 2 IR sensors
- Buzzer
- LED
- Breadboard or PCB
- Jumper wires
- Power source (USB or battery)

2. Sensor Placement

Fix the two IR sensors 13 cm apart in a straight line, ensuring they are properly aligned to detect an object passing in front of them.

3. Circuit Connections

- Connect the **IR sensor outputs** to two digital input pins on the Arduino Nano (e.g., D2 and D3).
- Connect the **buzzer** to a digital output pin (e.g., D4) through a current-limiting resistor.
- Connect the **LED** to another output pin (e.g., D5) with a resistor.
- Ensure all components share a **common ground (GND)** with the Arduino.

4. Code Uploading

- Write the Arduino sketch to:
 - Detect when each IR sensor is triggered.
 - Record the time (using `millis()` or `micros()`) when each sensor is triggered.
 - Calculate the speed using the time difference and the known distance (13 cm).
 - Activate the LED and buzzer if a certain speed threshold is exceeded (optional).
- Upload the code to the Arduino Nano using the Arduino IDE.

5. Power the System

Connect the Arduino Nano to a computer or battery to power the system.

6. Testing the System

- Pass an object in front of the sensors.
- Observe the serial monitor (or use LED/buzzer indicators) to see the calculated speed.
- Repeat with different speeds to test accuracy and consistency.

7. Troubleshooting & Calibration

- Adjust sensor sensitivity and placement if detection is inconsistent.
- Fine-tune the code for better timing precision and response.

8. Final Soldering (Optional)

- Once tested successfully on a breadboard, solder the components onto a PCB for a more permanent setup.

12. Observation

1. IR Sensor Response

- Both IR sensors successfully detected the object passing in front of them.
- The detection was more accurate when the object was close and moved straight across the sensor path.

2. Speed Calculation

- The speed was correctly calculated using the formula:
$$\text{Speed} = \text{Time} / \text{Distance} = 13 \text{ cm} / \text{Time between sensor triggers}$$
- Results varied slightly based on object size, lighting conditions, and speed.

3. Timing Accuracy

- The Arduino's `millis()` or `micros()` functions provided sufficient resolution for measuring time intervals.
- Speed values were more accurate for slower-moving objects due to better timing capture.

4. LED and Buzzer Output

- The buzzer and LED were triggered correctly when a speed threshold was crossed (if implemented in code).
- Output responses were immediate and consistent.

5. Environmental Factors

- Ambient light and reflections occasionally caused false triggers, especially under bright lighting conditions.

- Covering the IR sensor sides helped reduce external interference.

6. Overall System

- The system functioned as expected in multiple test runs.
- Minor delays or missed triggers were observed if the object moved too fast or wasn't aligned properly.



NETAJI SUBHASH UNIVERSITY OF TECHNOLOGY

Electronics Design Workshop

Project Report

Speed Detector System

ECECC402

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Academic Year : 2024 - 2025

1. Acknowledgement

We express our sincere gratitude to all those who have contributed to the completion of this project. It has been a journey marked by collaborative efforts, and we are deeply appreciative of the support and assistance we have received.

First and foremost, we extend our heartfelt thanks to our mentors, Prof. Dhananjay V. Gadre, Mr. Hiranya Kumar, and Mr. Amit Kumar Gupta, whose unwavering guidance and supervision have been instrumental throughout this endeavor. Their profound insights, constant support, and provision of essential information have been invaluable.

We are indebted to Prof. Gadre for his profound mentorship, which has illuminated our path and enriched our understanding of the project's intricacies. His willingness to share his extensive knowledge has been a beacon of inspiration, enabling us to navigate through the complexities of the project with confidence.

Furthermore, we express our appreciation to all the individuals who have extended their assistance and encouragement at various stages of this project. Their collective efforts have contributed significantly to its fruition, and for that, we are truly grateful.

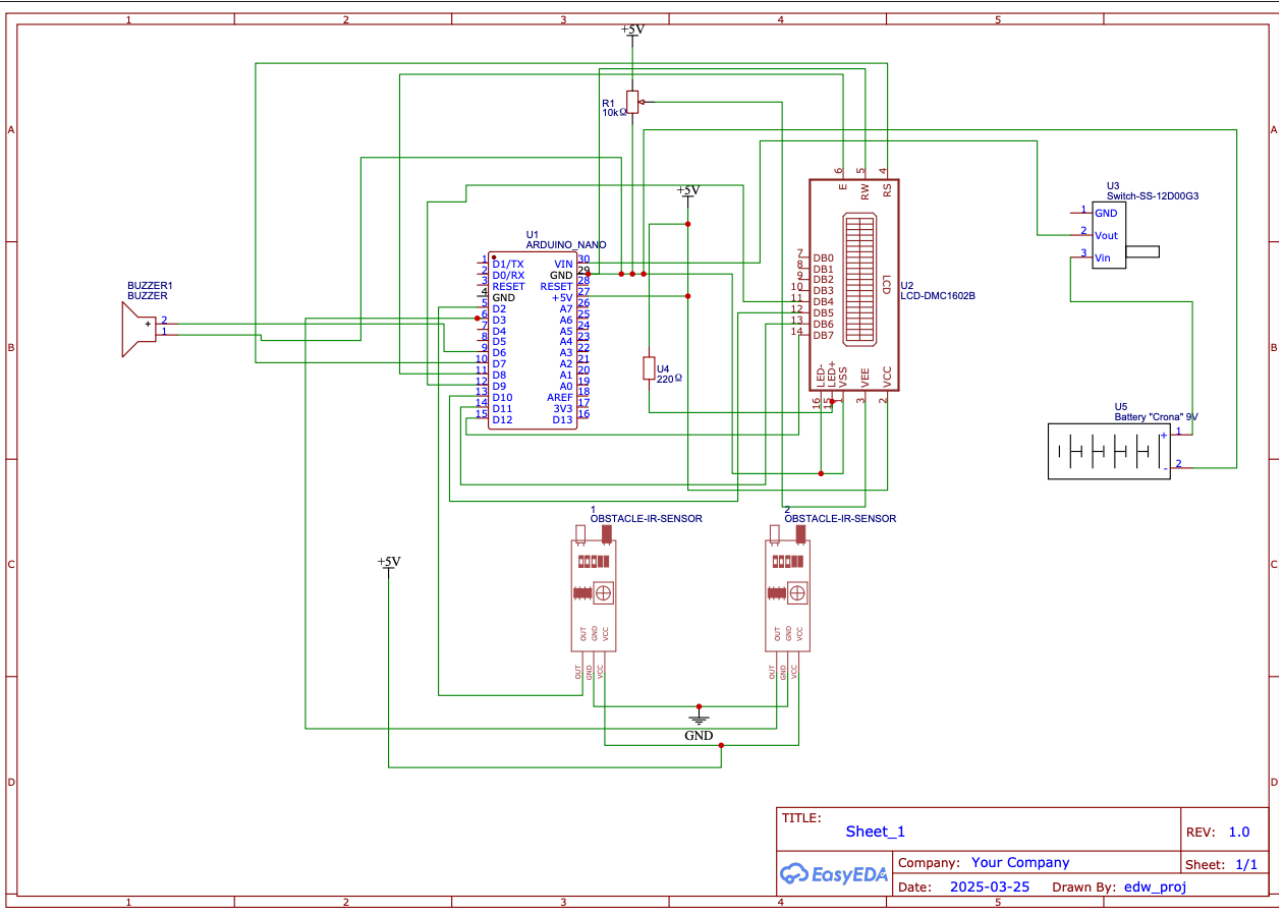
We also express our deepest gratitude to our parents for their unwavering love, encouragement, and sacrifices. Their support has been the bedrock of our journey, and we are profoundly thankful for their guidance and understanding.

2. Certificate

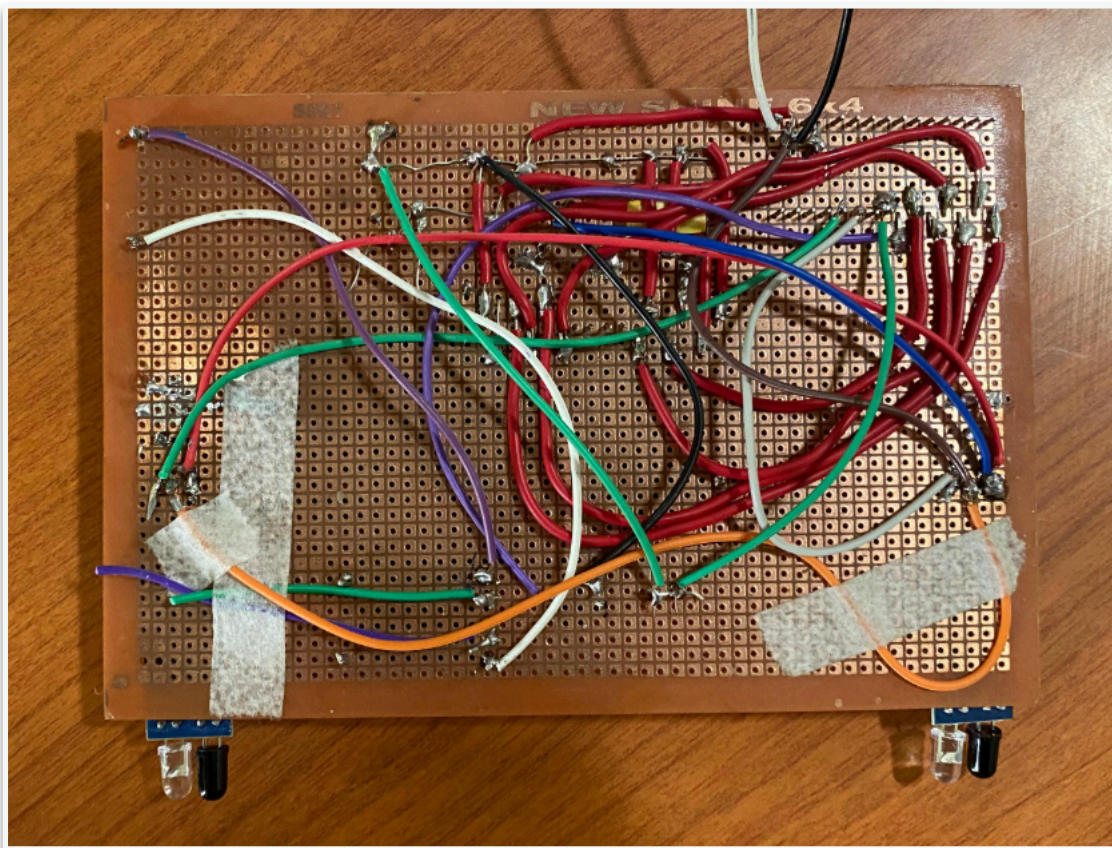
This is to certify that Pranjal Kumar (2023UEC2667), Advik Srivastava (2023UEC2677), and Ashutosh Kumar Singh (2023UEC2698) have successfully completed their ELECTRONICS DESIGN WORKSHOP project ‘Speed Detector System’ under the guidance of Prof. Dhananjay V. Gadre, Mr. Hiranya Kumar, and Mr. Amit Kumar Gupta.

S.No	Student Name	Student Roll No	Student Image	student Signature(date)
1.	Pranjal Kumar	2023UEC2667		
2.	Advik Srivastava	2023UEC2677		
3.	Ashutosh Kumar Singh	2023UEC2698		

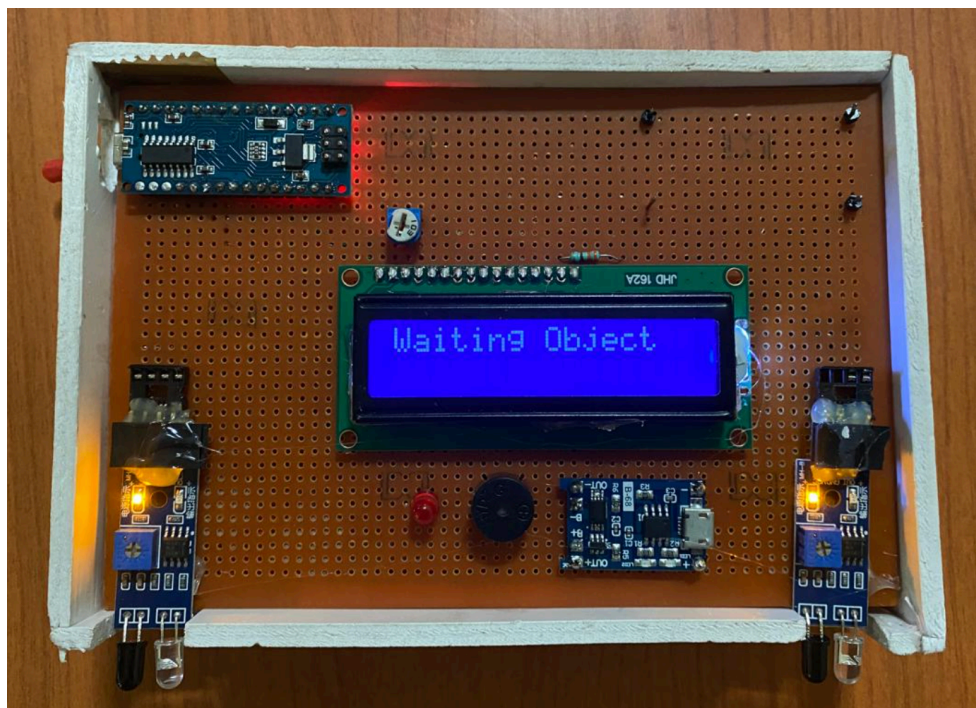
5. Circuit Diagram



6. PCB Design



7. Final Project Implementation



With Enclosure

